Sec: Sr. IIT **Date: 02-03-24**

Daily Practices Sheet - 08

MATHS

1. The number of rational terms in the expansion of $\left(1+\sqrt{2}+\sqrt[3]{3}\right)^6$ is

1) 6 2) 7 3) 5 4) 8

2. If the equations $a(y+z)=x, b(z+x)=y$ and $c(x+y)=z$, where $a \neq -1$, $b \neq -1, c \neq -1$, admit of nontrivial solutions then $(1+a)^{-1}+(1+b)^{-1}+(1+c)^{-1}$ is

1) 2 2) 1 3) $\frac{1}{2}$ 4)-2

3. If $A^2 = 8A + kI$ where $A = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix}$ then k is

1) 7 2) -7 3) 1 4) -1

4. If $ax^2+bx+6=0$ does not have two distinct real roots, where $a \in R$, $b \in R$, then the least value of $3a+b$ is

1) 4 2) -1 3) 1 4) -2

5. If n is an odd integer greater than or equal to 1 then the value of $n^3-(n-1)^3+(n-2)^3-....+(-1)^{n-1}.1^3$ is

1) $\frac{(n+1)^2.(2n-1)}{4}$ 2) $\frac{(n-1)^2.(2n-1)}{4}$

3) $\frac{(n+1)^2.(2n+1)}{4}$ 4) $\frac{(n-1)^2(2n+1)}{4}$

6. If a,b,c,d and p are distinct real numbers such that $\left(a^2+b^2+c^2\right)p^2-2(ab+bc+cd)p+\left(b^2+c^2+d^2\right)\leq 0$ then a,b,c,d are in

1) AP 2) GP 3) HP 4) AGP

7. The statement $p\rightarrow(q\rightarrow p)$ is equivalent to

1) $p\rightarrow q$ 2) $p\rightarrow(p\vee q)$ 3) $p\rightarrow(p\rightarrow q)$ 4) $p\rightarrow(p\wedge q)$

8. Of the members of three athletic teams in a school, 21 are in the cricket team, 26 are in the hockey team and 29 are in the foot ball team. Among them, 14 play hockey and cricket, 15 play hockey and football, and 12 play football and cricket. Eight play all the three games. The total number of members in the three athletic team is

1) 43 2) 76 3) 49 4) 61

9. If the number of 5-element subsets of the set $A\left\{a_1,a_2.......a_{20}\right\}$ of 20 distinct elements is k times the number of 5-element subsets containing a_4, then k is

1)5 2) $\frac{20}{7}$ 3) 4 4) $\frac{10}{3}$

10. The number of divisors of the form $4n+2(n\geq 0)$of the integer 240 is

1) 4 2) 8 3) 10 4) 3

11. A fair coin is tossed repeatedly. The probability of getting a result in the fifth toss different from those obtained in the first four tosses is

1) $\frac{1}{2}$ 2) $\frac{1}{32}$ 3) $\frac{31}{32}$ 4) $\frac{1}{16}$

12. The mean of the numbers a,b,8,5,10 is 6 and the variance is 6.80 .Then which one of the following gives possible values of a and b

1) a=0, b=7 2) a=5, b=2 3) a=1,b=6 4) a=3,b=4

13. Let $\vec{a}=2\vec{i}-\vec{j}+\vec{k}$, $\vec{b}=\vec{i}+2\vec{j}-\vec{k}$ and $\vec{c}=\vec{i}+\vec{j}-2\vec{k}$. A vector in the plane of $\vec{b}$ and $\vec{c}$ whose projection on $\vec{a}$ has the magnitude $\sqrt{2/3}$ is

1) $2\vec{i}+3\vec{j}-3\vec{k}$
2) $2\vec{i}+3\vec{j}+3\vec{k}$
3) $-2\vec{i}-\vec{j}+5\vec{k}$
4) $2\vec{i}+\vec{j}+5\vec{k}$

14. In the ΔABC, the coordinates of B are (0,0), AB=2, $\angle ABC=\frac{\pi}{3}$ and the middle point of BC has the coordinates (2,0). The centroid of the triangle is

1) $\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right)$
2) $\left(\frac{5}{3},\frac{1}{\sqrt{3}}\right)$
3) $\left(\frac{4+\sqrt{3}}{3},\frac{1}{3}\right)$
4) $\left(\frac{1}{2},\frac{1}{\sqrt{3}}\right)$

15. If $P\left(1+\frac{t}{\sqrt{2}},2+\frac{t}{\sqrt{2}}\right)$ be any point on a line then the range of values of t for which the point P lies between the parallel lines x+2y=1 and 2x+4y=15 is

1) $-\frac{4\sqrt{2}}{3}<t<\frac{5\sqrt{2}}{6}$
2) $0<t<\frac{5\sqrt{2}}{6}$
3) $-\frac{4\sqrt{2}}{5}<t<0$
4) $\frac{-5\sqrt{2}}{6}<t<\frac{4\sqrt{2}}{5}$

16. Two distinct chords drawn from the point (p,q) on the circle $x^2+y^2=px+qy$, where $pq\neq 0$, are bisected by the x-axis. Then

1) $|p|=|q|$
2) $p^2=8q^2$
3) $p^2<8q^2$
4) $p^2>8q^2$

17. The ends of a line segment are P(1,3) and Q(1,1). R is a point on the line segment PQ such that R divides $\overline{PQ}$ internally in the ratio $1:\lambda$. If R is an interior point of the parabola $y^2=4x$ then

1) $\lambda\in(0,1)$
2) $\lambda\in\left(-\frac{3}{5},1\right)$
3) $\lambda\in\left(\frac{1}{2},\frac{3}{5}\right)$
4) $\lambda\in\left(-\frac{3}{5},\frac{3}{5}\right)$

18. P is a variable point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=2$ (a>b)whose foci are F_1 and F_2. The maximum area (in unit[2]) of the $\Delta PFF'$ is

1) $2b\sqrt{a^2-b^2}$ 2) $\sqrt{2}b\sqrt{a^2-b^2}$ 3) $b\sqrt{a^2-b^2}$ 4) $2a\sqrt{a^2-b^2}$

19. The locus of a point $P(\alpha,\beta)$ moving under the condition that the line $y=\alpha x+\beta$ is a tangent to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is

1) a circle
2) an ellipse
3) a hyperbola
4) a parabola

20. ABC is a triangle where $A=(2,3,5), B=(-1,3,2)$ and $C=(\lambda,5,\mu)$. If the median through A is equally inclined with the axes then

1) $\lambda=14,\mu=20$ 2) $\lambda=7,\mu=10$ 3) $\lambda=\frac{7}{2},\mu=5$ 4) $\lambda=10,\mu=7$

21. The shortest distance between the line $\frac{x-3}{3}=\frac{y}{0}=\frac{z}{-4}$ and the y-axis is

1) $\frac{1}{5}$ 2) 1 3) 0 4) $\frac{12}{5}$

22. Let $f(x)=\begin{cases}\sin x, x\neq n\pi \\ 2, x=n\pi, \text{where } n\in Z\end{cases}$ and $g(x)=\begin{cases}x^2+1, x\neq 2 \\ 3, x=2\end{cases}$. Then $\lim\limits_{x\to 0} g(f(x))$ is

1) 0 2) 1 3) 3 4) 2

23. Let $f(x)=[\cos x+\sin x], 0<x<2\pi$ where [x] denotes the greatest integer less than or equal to x. the number of points of discontinuity of f(x) is

1) 6 2) 5 3) 4 4) 3

24. If $f(x)=(ab-b^2-2)x+\int_0^x(\cos^4\theta+\sin^4\theta)d\theta$ is a decreasing function of x for all $x\in R$ and $b\in R$, b being independent of x, then

1) $a\in(0,\sqrt{6})$ 2) $a\in(-\sqrt{6},\sqrt{6})$ 3) $a\in(-\sqrt{6},0)$ 4) $a\in(-6,6)$

25. $\int \frac{dx}{x^{1/5}\left(1+x^{4/5}\right)^{1/2}}$ is

1) $\sqrt{1+x^{4/5}}+k$ 2) $\frac{5}{2}\sqrt{1+x^{4/5}}+k$

3) $x^{4/5}\left(1+x^{4/5}\right)^{1/2}+k$ 4) $x^{1/5}\left(1+x^{1/5}\right)^{1/2}+k$

26. If $f(x)$ be a quadratic polynomial such that $f(0)=2, f'(0)=-3$ and $f''(0)=4$ then $\int_{-1}^{1} f(x)dx$ is equal to

1) -3 2) 16/3 3) 0 4) -16/3

27. The ordinate x=a divides the area bounded by the x-axis, the curve $y=1+\frac{8}{x^2}$ and the ordinates x=2 and x=4 into equal parts. Then a is

1) 3 2) $\frac{7}{2}$ 3) $2\sqrt{2}$ 4) $\frac{5}{2}$

28. The solution of theequation $\left(x^2+y^2\right)dy = xy.dx$ is $y=y(x)$. If $y(1)=1$ and $y(x_0)=e$ then x_0 is

1) $\sqrt{2\left(e^2-1\right)}$ 2) $\sqrt{2\left(e^2+1\right)}$ 3) $\sqrt{3}e$ 4) $\sqrt{\frac{1}{2}\left(e^2+1\right)}$

29. The most general values of x for which $\sin x+\cos x = \min\limits_{a\in R}\left\{1, a^2-4a+6\right\}$ are given by

1) $2n\pi; n\in z$ 2) $2n\pi+\frac{\pi}{2}; n\in z$

3) $n\pi+(-1)^n.\frac{\pi}{4}-\frac{\pi}{4}; n\in z$ 4) $2n\pi-\frac{\pi}{2}; n\in z$

30. A flagstaff stands vertically on a pillar, the height of the flagstaff being double the height of the pillar. A man on the ground at a distance finds that both the pillar and the flagstaff subtend equal angles at his eyes. The ratio of the height of the pillar and the distance of the man from the pillar, is

1) $\sqrt{3}:1$ 2) $1:3$ 3) $1:\sqrt{3}$ 4) $\sqrt{3}:2$

PHYSICS

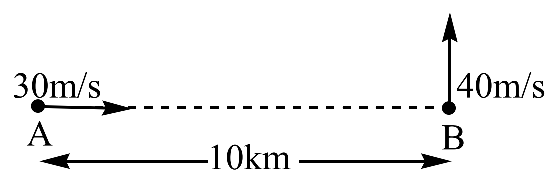
31. A physical quantity Q is related to four observables *x,y,z* and *t* by the relation

$$Q = \frac{x^{2/5} z^3}{y\sqrt{t}}$$

The percentage errors of measurement in *x,y,z* and *t* are 2.5%, 2%, 0.5% and 1% respectively. The percentage error in Q will be

1) 5%   2) 4.5%   3) 8%   4) 7.75%

32. Ship A is moving with velocity 30m/s due east and ship B with velocity 40m/s due north. Initial separation between the ships is 10km as shown in figure. After what time ships are closest to each other?

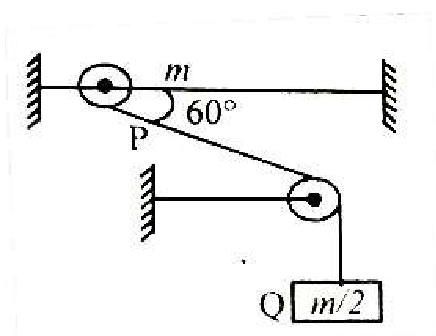


1) 80s   2) 120s   3) 160s   4) 100s

33. The greatest range of a particle, projected up with a given velocity on an inclined plane, is x times the greatest vertical altitude above the inclined plane in that condition. Find the value of x.

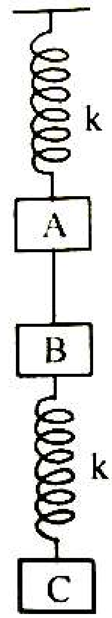
1) 2   2) 4   3) 3   4) ½

34. A smooth ring P of mass m can slide on a fixed smooth horizontal rod. A string tied to the ring passes over a fixed pulley and carries a block Q of mass (m/2) as shown in the fig. at an instant, the string between the ring and the pulley makes an angle 60º with the rod. The initial acceleration of the ring , if the system is released from rest is



1) $\frac{2g}{3}$ 2) $\frac{2g}{6}$ 3) $\frac{2g}{9}$ 4) $\frac{g}{3}$

35. The system shown in the given fig is in equilibrium. All the blocks are of the equal mass m and springs are ideal. When the string between A and B is cut then immediately after that



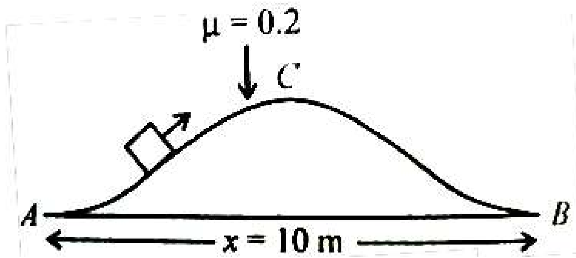
1) the acceleration of block A is 2g while the acceleration of block C is g

2) the acceleration of block B is 2g while the acceleration of block C is zero

3) the acceleration of block A is 2g while the acceleration of block B is g

4) the acceleration of block B is g while the acceleration of block C is zero

36. A block of mass 1kg is pulled slowly along the curved path ACB by a tangential force as shown in figure. The work done against the frictional force when the block moves from A to B is

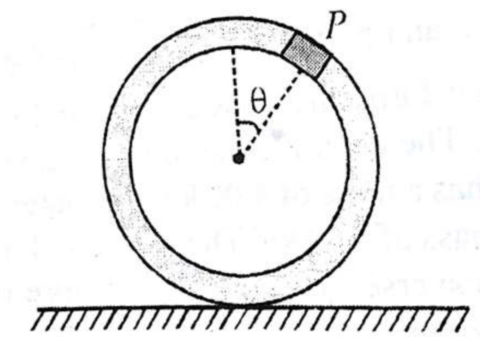


1) 5J 2) 10J 3) 20J 4) -20 J

37. A block of mass m=1kg is moving with a constant acceleration $a = 1ms^{-2}$ on a rough horizontal plane. The coefficient of friction between the block and plane is $\mu = 0.1$. The initial velocity of block is zero. The power delivered by the external agent at a time t=2s from the beginning is equal to (take g=$10ms^{-2}$)

1) 1 W 2) 2W 3) 3W 4) 4W

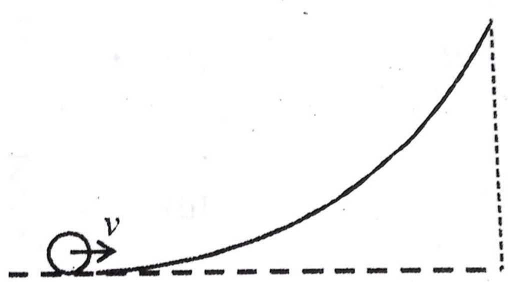
38. A small block of mass 'm' is rigidly attached at 'P' to a ring of mass '3m' and radius 'r'. The system is released from rest at $\theta = 90^{o}$ and it starts rolling without sliding.



The angular acceleration of hoop just after release is

1) g/4r 2) g/8r 3) g/3r 4) g/2r

39. A small object of uniform density rolls up a curved surface with an initial velocity *v*. It reaches up to a maximum height of $\frac{3v^2}{4g}$ with respect to the initial position. The object is



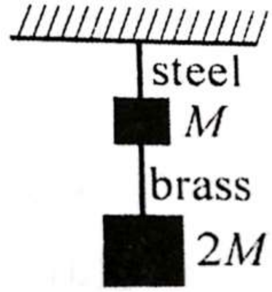
1) ring 2) solid sphere
3) hollow sphere 4) disc

40. The minimum and maximum distances of a satellite from centre of earth are 2R and 4R respectively, where R is the radius of earth. The minimum and maximum speeds of the satellite will be(M = mass of the earth)

1) $\sqrt{\frac{GM}{R}}, \sqrt{\frac{2GM}{R}}$ 2) $\sqrt{\frac{GM}{6R}}, \sqrt{\frac{2GM}{3R}}$

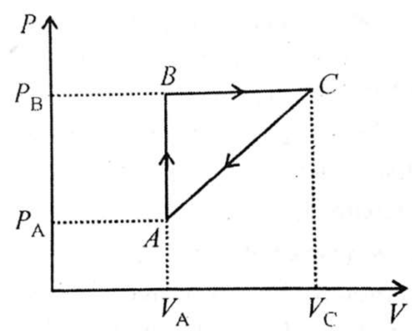
3) $\sqrt{\frac{2GM}{3R}}, \sqrt{\frac{4GM}{3R}}$ 4) $\sqrt{\frac{GM}{6R}}, \sqrt{\frac{GM}{3R}}$

41. If the ratio of lengths, radii and Young's moduli of steel and brass wires in the figure are *a,b* and *c* respectively, then the corresponding ratio of increase in their lengths is



1) $2a^2cb$ 2) $3a/2b^2c$ 3) $2ac/b^2$ 4) $3c/2ab^2$

42. A thermodynamical process is shown in the fig with $P_A = 3\times P_{atm}$, $V_A = 2\times10^{-4}m^3, P_B = 8\times P_{atm}, V_C = 5\times10^{-4}m^3$, in the process AB and BC, 600J and 200J heat are added to the system, respectively. Find the change in internal energy of the system in the process CA. $\left(1P_{atm} = 10^5 N/m^2\right)$

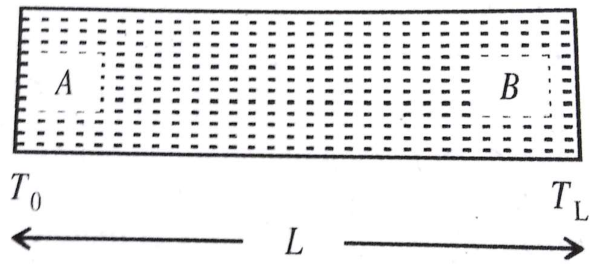


1) 560J 2) -560J 3) -40J 4) +40J

43. The relation between internal energy U, pressure P and volume V of an ideal gas in an adiabatic process is U=2+3PV. What is the value of the ratio of the molar specific heats $\left(\frac{C_P}{C_V}\right)$?

1) 2/3 2) 4/3 3) 3/2 4) 1

44. The temperature of mono-atomic gas in a uniform container of length L varies linearly from T_0 to T_L as shown in the figure. If the molecular weight of the gas is M, then the time taken by a wave pulse in travelling from end A to end B is



1) $\frac{2L}{\left(\sqrt{T_L}+\sqrt{T_0}\right)}\sqrt{\frac{3M}{5R}}$ 2) $\sqrt{\frac{3\left(T_L-T0\right)}{5RML}}$

3) $\frac{L}{\left(\sqrt{T_L}-\sqrt{T_0}\right)}\sqrt{\frac{3M}{5R}}$ 4) $\sqrt{\frac{M}{2R\left(T_L-T_0\right)}}$

45. Wavelength of red light is $7.5\times10^{-5}\,cm$ and that of blue light is 5×10^{-5} cm. In Young's experiment, the value of *n* for which $\left(n+1\right)^{th}$ the blue bright band coincides with n^{th} red bright band is

1) 8 2) 4 3) 2 4) 1

46. In Young's double-slit experiment, the separation between the slits is halved and the distance between the slits and the screen is doubled. The fringe width is

1) unchanged 2) halved 3) doubled 4) quadrupled

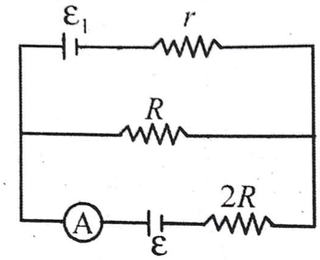
47. A capacitor is charged to a potential difference of 100V and is then connected across a resistor. The potential difference across the capacitor decays exponentially with respect to time. After 1s, the potential difference between the plates of the capacitor is 80V. After 2s, the potential difference between the plates will be

1) 40V 2) 56V 3) 60V 4) 64V

48. If the electrostatic potential varies in the space as $\phi=\phi_0\left(x^2+y^2+z^2\right)$ where ϕ_0 is constant, then the charge density giving rise to the above potential would be

1) 0 2) $-6\phi_0\varepsilon_0$ 3) $-2\phi_0\varepsilon_0$ 5) $-\frac{6\phi_0}{\varepsilon_0}$

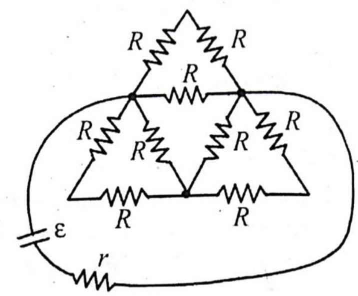
49. The incorrect option in the given circuit if the reading of ammeter is zero is



1) The value ε_1 will be $\dfrac{\varepsilon(R+r)}{R}$ 2) Current in R is $\dfrac{\varepsilon_1}{r+R}$

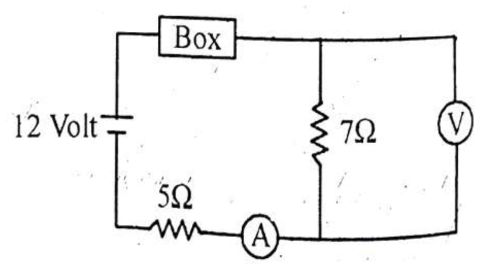
3) Value of ε_1 will be ε 4) Potential across 2R is zero

50. For maximum power delivered to the external circuit by the battery, the internal resistance of battery r is



1) 10R 2) $\dfrac{4R}{9}$ 3) $\dfrac{R}{8}$ 4) $\dfrac{10R}{9}$

51. As shown in circuit diagram, an ideal voltmeter and an ideal ammeter readings are zero. Sum of potential drops across box and 5Ω resistance will be (given that there are no faults in voltmeter and ammeter)



1) 7volt 2) 5volt 3) 12volt 4) data insufficient

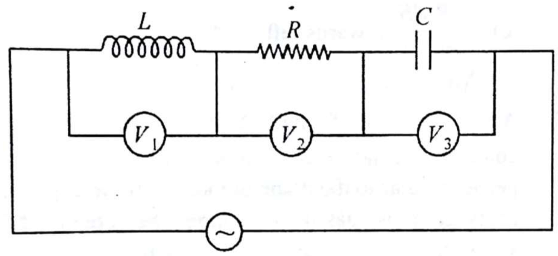
52. A long straight wire along the z-axis carries a current I in the negative z direction. The magnetic vector field $\vec{B}$ at a point having coordinates *(x,y)* in the z=0 plane is

1) $\dfrac{\mu_0 I\left(y\hat{i}-x\hat{j}\right)}{2\pi\left(x^2+y^2\right)}$ 2) $\dfrac{\mu_0 I\left(x\hat{i}+y\hat{j}\right)}{2\pi\left(x^2+y^2\right)}$ 3) $\dfrac{\mu_0 I\left(x\hat{j}-y\hat{i}\right)}{2\pi\left(x^2+y^2\right)}$ 4) $\dfrac{\mu_0 I\left(x\hat{i}-y\hat{j}\right)}{2\pi\left(x^2+y^2\right)}$

53. The current in an L-R circuit builds up to (3/4)th of its steady state value in 4 seconds. The time constant of this circuit is

1) $\frac{1}{\ln 2}s$ 2) $\frac{2}{\ln 2}s$ 3) $\frac{3}{\ln 2}s$ 4) $\frac{4}{\ln 2}s$

54. If the readings of V_1 and V_3 are 100volt each, then reading of V_2 is



1) 0volt
2) 100volt
3) 200volt
4) cannot be determined by given information

55. A red LED emits at 0.1 watt uniformly around. The ratio of amplitude of electric field of light at a distance of 1 m from the diode to the field at a distance 4 m from the diode is

1) 4 2) 2.45 3) 16 4) 2

56. The first member of Balmer series of hydrogen has a wavelength of $6563\overset{0}{\text{A}}$. The wavelength of its second member will be

1) $4861\overset{0}{\text{A}}$ 2) $6563\overset{0}{\text{A}}$ 3) $3561\overset{0}{\text{A}}$ 4) $1215\overset{0}{\text{A}}$

57. If the potential energy in ground state of a hydrogen atom is considered to be zero, then find the total energy is first excited state

1) 27.2 eV 2) 23.8 eV 3) 13.6 eV 4) 6.8 eV

58. Find the number of photons emitted per second by a 25 watt source of monochromatic light of wavelength $6600\overset{0}{\text{A}}$. What is the photoelectric current assuming 3% efficiency for photoelectric effect?

1) $\frac{25}{3}\times 10^{19} J, 0.4$ A 2) $\frac{25}{4}\times 10^{19} J, 6.2$ A

3) $\frac{25}{2}\times10^{19}J,0.5$ A   4) $\frac{25}{3}\times10^{19}J,4$ A

59. In a forward biased P-N junction carrying a certain amount of current, which of the following statement is false?

1) The depletion region width decreases in comparison to unbiased case

2) The electric field strength varies in depletion region

3) The neutral regions behave like ohmic resisters with small resistance.

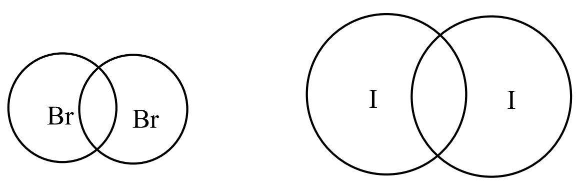
4) The charge density in the depletion region is uniform.

60. A TV transmission tower has a height of 240 m. Signals broadcast from this tower will not be received by LOS communication at a distance of (assume theradius of earth to be $6.4\times10^{6}m$)

1) 100 km   2) 24 km   3) 55 km   4) 50 km

CHEMISTRY

61. The below diagram shows molecules of Br_2 and I_2 draw to the nearly same scale. Which of the following is the best explanation for the difference in the boiling points of liquid Br_2 and I_2, which are 59ºC and 184ºC respectively



1) Solid iodine is a network covalent solid where as solid bromine is a molecular solid

2) The covalent bonds in I_2 molecules are weaker than those in Br_2 molecules

3) I_2 molecules have electron clouds that are more polarizable than those of Br_2 molecules, thus London dispersion forces are stronger in liquid I_2

4) Bromine has a greater electronegativity than iodine, thus there are stronger dipole – dipole forces in liquid bromine than in liquid iodine

62. Lewis electron – dot diagrams for CO_2 and SO_2 are given below. The molecular geometry and polarity of the two substances are

1) The same because the molecular formulas are similar

2) The same because carbon and sulphur have similar electronegativity values (2.5)

3) Different because the lone pair of electrons on the S atom make it the negative end of a dipole

4) Different because S atom has greater number of electron domains (regions of electron density) surrounding it than carbon has

63. Solid NaCl melts at a temperature of 800ºC while solids NaBr melts at 750ºC. Which of the following is an explanation for the higher melting point of NaCl.

1) Chloride ion has less mass than a bromide ion

2) Chloride ion has greater negative charge than a bromide ion

3) Chloride ion is smaller than a bromide ion

4) Chloride ion has less number of protons than bromide ion in their nuclei.

64. Solubility of NaCl in heavy water is

1) Same as that in H_2O 2) 15% lower than that in H_2O

3) 15% more than that in H_2O 4) 100% more than that in H_2O

65. In medieval European cathedral(in cold countries) the people found that organ pipes become brittle end powdered, where it was believed to be due to the Devil's work. The reason for this Devil's work is

1) Devils destroyed them with angry on the god.

2) It is devils charge on cathedral people

3) It is due to change in one crystalline allotrope to another crystalline allotrope

4) It is due to change in bonding between the metal atoms.

66. The ions of oxoacids of halogens are stabilized due to

1) $p\pi - d\pi$ bonding between 2p orbitals on oxygen with d orbitals on the halogen atom and due to resonance

2) $p\pi - d\pi$ bonding between full p orbitals on the halogen atom and resonance

3) $p\pi - d\pi$ bonding between full 2p orbitals on oxygen with p orbitals on the halogen atom

4) $d\pi - d\pi$ bonding between d orbitals on oxygen with full 2p orbitals on the halogen atom

67. H_2S is less stable than H_2O because

1) the bonding orbitals of sulphur are larger and more diffused than those of oxygen, and hence they overlap less effectively with the 1s orbital of the hydrogen atom.

2) the bonding orbitals of sulphur are smaller and more diffused than those of oxygen, and hence they overlap less effectively with the 1s orbital of the hydrogen atom.

3) the bonding orbitals of sulphur are smaller and less diffused than those of oxygen, and hence they overlap less effectively with the 1s orbital of the hydrogen atom

4) H_2O molecules form H bonds but H_2S molecules do not

68. Cis platinum complexes are

1) used in the extraction of silver and gold

2) effective antidotes for heavy metal poisoning (eg. Pb^{2+} and Hg^{2+})

3) used to prevent eutrophication of lakes

4) effective antitumor agent

69. For the following two compounds

Compound – 1 : $\left[Cr\left(H_2O\right)_6\right]Cl_3$

Compound – 2 : $\left[Cr\left(NH_3\right)_6\right]Cl_3$

1) Compound 1 is violet / compound 2 is yellow

2) Compound 1 is yellow/ compound 2 is Violet

3) Both compounds are colourless

4) Both compounds have same colour

70. Weed, insect, and fungus pests can be controlled by applying specific pesticides called (in the correct order)

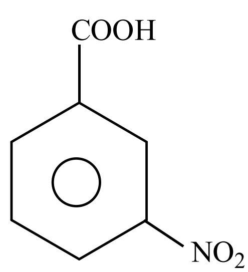
1)DDT, dieldrin, and fungicides.

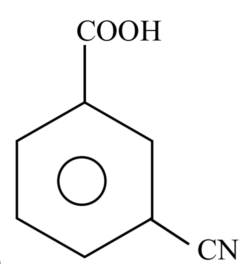
2)herbicides, and insecticides.

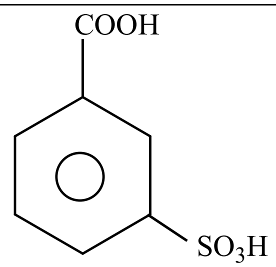
3)fungicides, herbicides, and insecticides.

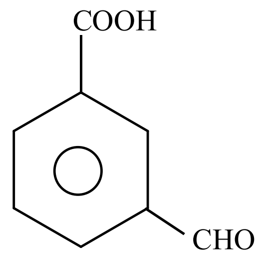
4) herbicides, insecticides, and fungicides

71. Which of the following molecule is strongest acid

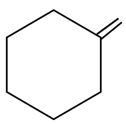
1) 

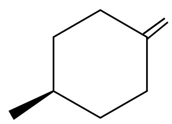
2) 

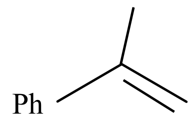
3) 

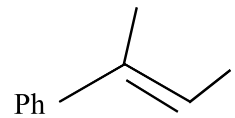
4) 

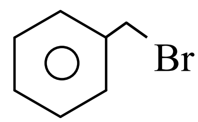
72. Which of the following alkene will give enantiomeric product on reaction with HCl

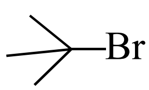
1) 

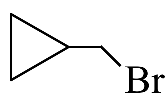
2) 

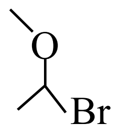
3) 

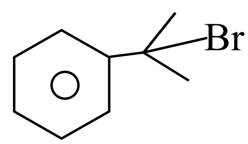
4) 

73. Which of the following halides are more reactive than  toward SN^1


I

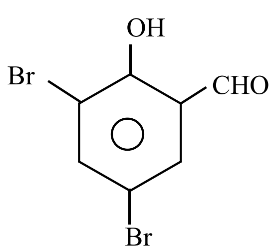

II

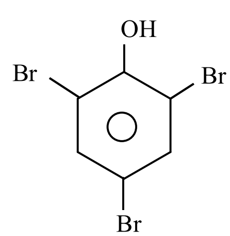

III

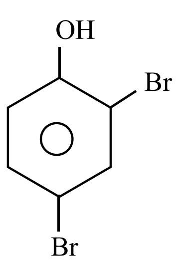

IV

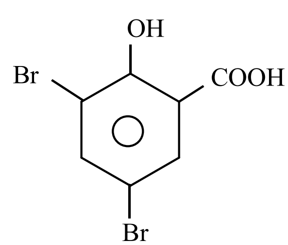
1) Only I, II 2) Only I, II, III 3) Only I, II, IV 4) I, II, III, IV

74. Phenol on treatment with $CHCl_3$ in the presence of NaOH followed by acidification produces compound X as the major product. X on treatment with $KMnO_4 + H_2SO_4$ followed by $Br_2 + H_2O$ give Y. Then 'Y' is

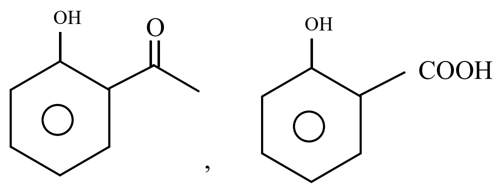
1) 

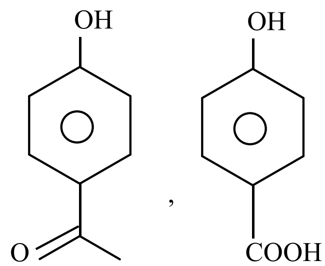
2) 

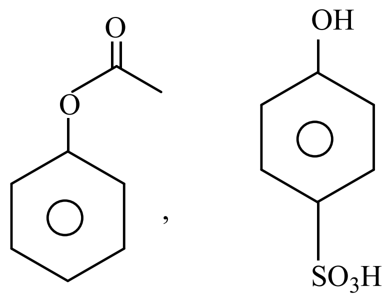
3) 

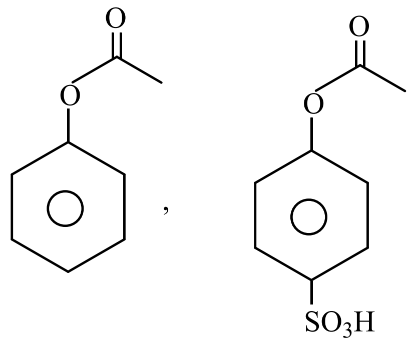
4) 

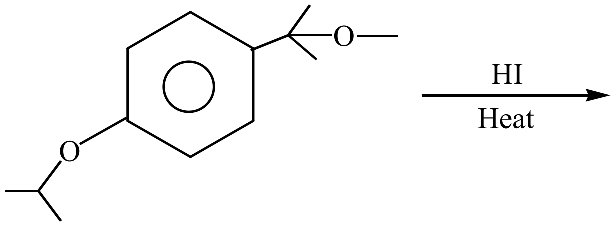
75. Phenol reacts with acetyl chloride in the presence of NaOH to form product A. 'A' reacts H_2SO_4 to form product B. A and B are respectively.

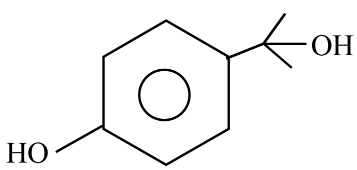
1) 

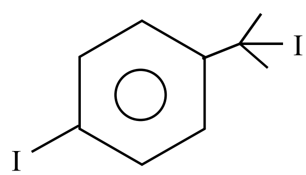
2) 

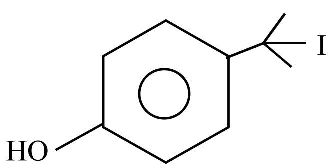
3) 

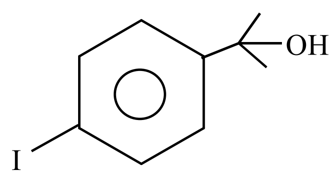
4) 

76. 

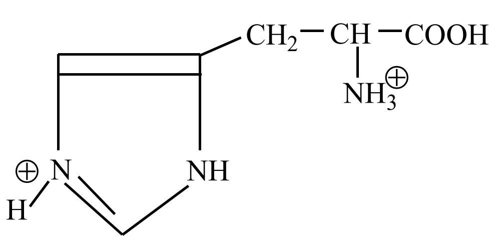
1) 

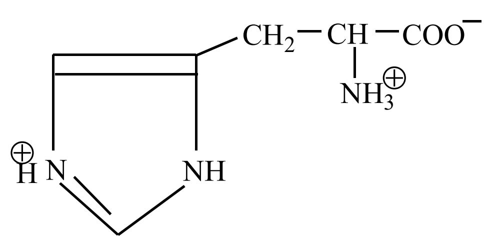
2) 

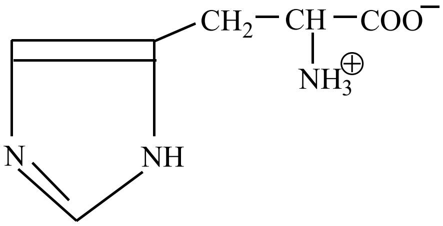
3) 

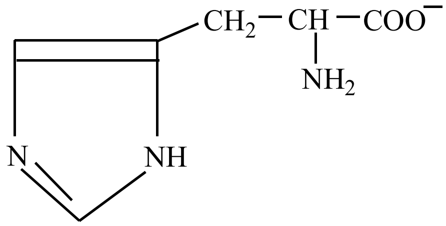
4) 

77. The predominant form of histidine present in pH = 8 solution.

1) 

2) 

3) 

4) 

78. D-glucose on oxidation with con HNO_3 gives

1) D-gluconic acid

2) 2,3,4,5,6 – pentahydroxy hexanoic acid

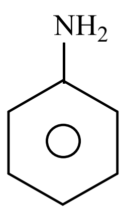
3) D-aldonic acid

4) D – Glucaric acid

79. Which of the following nitrogen-containing compounds does not respond positively to Lassaigne's test for nitrogen

1) $NH_2-\overset{\overset{\large O}{||}}{C}-NH-NH_2$

2) $NH_2-\overset{\overset{\large O}{||}}{C}-NH_2$

3) 

4) NH_2-NH_2

80. A match box exhibits ______ geometry

1) Cubic
2) Orthorhombic
3) Triclinic
4) Monoclinic

81. The process of getting freshwater from seawater is known as

1) Osmosis
2) Filtration
3) Pressure distillation
4) Reverse osmosis

82. 20ml of a mixture of oxygen and ozone was heated till ozone is completely decomposed. The mixture on cooling was found to have a volume of 21ml. The volume percentage of ozone in the initial gaseous mixture is

1) 20%
2) 10%
3) 80%
4) 30%

83. At certain temperature pH of 10^{-2} M NaOH is 10. Then the pH of neutral solution is

1) 5
2) 6
3) 7
4) 6.5

84. An aqueous solution of potash alum is acidic in nature. This is due to the hydrolysis of ____ ions

1) K^{+} 2) Al^{+3} 3) SO_4^{2-} 4) All of the above

85. Calculate the minimum mass of NaCl necessary to dissolve 0.01 mol of AgCl in 100L solution(Assume no change in volume)

$K_f\,AgCl_2^{-} = 3\times10^{5}$

$K_{sp}\,AgCl = 10^{-10}$

1) 19.3 kg 2) 193 kg 3) 1.93 kg 4) 16kg

86. pH of water is 7.0 at 25ºC. If water is heated to 70ºC, the

1) pH will decrease and the sample becomes acidic.

2) pH will increase but the sample will remain neutral.

3) pH will remain constant as 7.

4) pH will decrease but the sample will remain neutral.

87. Consider the following equilibrium in a closed container:

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

At a fixed temperature, the volume of the reaction container is halved. For this change, which of the following statements holds true regarding the equilibrium constant (K_p) and degree of dissociation (α)?

1) neither K_p nor α changes

2) both K_p and α changes

3) K_p changes, but α does not change

4) K_p does not change, but α changes

88. Which of the following statement is correct?

1) E_{Cell} and $\Delta_r G$ of cell reaction both are extensive properties.

2) E_{Cell} and $\Delta_r G$ of cell reaction both are intensive properties

3) E_{Cell} is an intensive property while $\Delta_r G$ of cell reaction is an extensive property.

4) E_{Cell} is an extensive property while $\Delta_r G$ of cell reaction is an intensive property.

89. Which of the following statements is incorrect about the collision theory of chemical reaction?

1) It considers reacting molecules or atoms to be hard spheres and ignores their structural features

2) Number of effective collisions determines the rate of reaction.

3) Collision of atoms or molecules possessing sufficient threshold energy results into the product formation

4) Molecules should collide with sufficient threshold energy and proper orientation for the collision to be effective.

90. Maximum work can a gas do, if it is allow to expand isothermally against

1) vacuum   2) high pressure of surrounding

3) low pressure of surrounding   4) atmospheric pressure

KEY SHEET

1	**2**	2	**1**	3	**2**	4	**4**	5	**1**
6	**2**	7	**2**	8	**1**	9	**3**	10	**1**
11	**4**	12	**4**	13	**3**	14	**2**	15	**1**
16	**4**	17	**1**	18	**1**	19	**3**	20	**2**
21	**4**	22	**2**	23	**3**	24	**2**	25	**2**
26	**2**	27	**3**	28	**3**	29	**3**	30	**3**
31	**1**	32	**2**	33	**2**	34	**3**	35	**2**
36	**3**	37	**4**	38	**2**	39	**4**	40	**2**
41	**2**	42	**2**	43	**2**	44	**1**	45	**3**
46	**4**	47	**4**	48	**2**	49	**3**	50	**2**
51	**3**	52	**1**	53	**2**	54	**3**	55	**1**
56	**1**	57	**2**	58	**1**	59	**4**	60	**3**
61	**3**	62	**4**	63	**3**	64	**2**	65	**3**
66	**1**	67	**1**	68	**4**	69	**1**	70	**4**
71	**3**	72	**4**	73	**4**	74	**2**	75	**4**
76	**3**	77	**3**	78	**4**	79	**4**	80	**2**
81	**4**	82	**2**	83	**2**	84	**2**	85	**1**
86	**4**	87	**4**	88	**3**	89	**3**	90	**2**